When Regional Favoritism Fades: Nightlights and Pollution in Leader Birth Regions

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Compiled on April 21, 2026

We study whether national leaders direct cleaner economic growth to their birth regions rather than simply more growth. We define "green favoritism" as lower pollution intensity per unit of economic activity in politically favored regions. We combine the Political Leaders' Affiliation Database, ADM2-level nightlights, ACAG surface NO_2, ACAG $PM_{2.5}$, V-Dem democracy scores, and GADM administrative boundaries in a global district-year panel with region fixed effects, country-year fixed effects, and leader-period clustering. In the long DMSP replication window, we recover the classic birth-region nightlights result: the lag-0 coefficient is about 0.0128 in 1992–2013 ($p = 0.030$), remains positive in 1995–2013 and 2000–2013, and then collapses in 2005–2013 (0.0017, $p = 0.768$). The harmonized 2005–2019 nightlights coefficient is also essentially zero (0.0004, $p = 0.966$). A targeted search across geography, tenure, autocracy, luminosity headroom, and pollution-support subsamples finds that Africa and Sub-Saharan Africa drive a strong historical DMSP result, but no post-2005 subsample clears a credible activity-effect criterion. In the later pollution window, the lag-0 NO_2 coefficient is positive (0.0308, $p = 0.045$) and the pollution-intensity outcome log(NO_2) minus log(nightlights) is also positive (0.0333, $p = 0.063$). The evidence therefore does not support green favoritism in the post-2005 pollution window; if anything, the results suggest dirtier growth in birth regions. The main empirical development is attenuation of the nightlights reduced-form effect in later years.

regional favoritism | political economy | nighttime lights | air pollution | autocracy

Political leaders often direct resources toward their home regions, a pattern documented in cross-country evidence using subnational nightlights (1). Recent replication work suggests that the classic result remains visible in modern data, but also highlights the importance of data choice and periodization (2). This paper extends that literature by asking whether politically favored regions receive not only more development, but cleaner development. While Bora (2025) focus on nightlights replication and crisis dynamics, we contribute by extending the analysis to satellite-based pollution outcomes and diagnostic checks for post-2005 attenuation.

The hypothesis of "green favoritism" is straightforward: if leaders have the motive and capacity to protect their home regions, favoritism may operate through the composition of growth. Leaders might use their discretion over industrial siting, zoning, or public investment to direct cleaner sectors and amenity-enhancing projects toward birth districts while pushing pollution-intensive activity toward politically peripheral areas. That possibility matters for environmental inequality as much as for distributive politics (3).

The evidence points to a different contribution. The long-run replication still works, but the nightlights activity reduced form weakens sharply in the post-2005 period that overlaps with the available NO_2 and harmonized nightlights data. Because the later pollution analysis depends on a credible later-period favoritism effect, the central empirical task becomes explaining that attenuation rather than claiming definitive evidence of green favoritism. We therefore estimate targeted subsample tests in settings where the activity effect should be most likely to survive before interpreting any pollution outcomes.

1. Data and Empirical Design

The analysis combines five core inputs. First, the Political Leaders' Affiliation Database (PLAD) provides leader identity, spell timing, and subnational birthplace information used to code treatment (4). Second, nightlights provide the economic-activity proxy. We use a replication-style DMSP ADM2 panel for 1992–2013 and a harmonized DMSP-to-VIIRS panel for 2005–2019, with DMSP-based observations through 2013 and simulated VIIRS afterward (5). DMSP records luminance as an integer Digital Number (DN) from 0 to 63, with known saturation near the ceiling (6). Third, we use ACAG surface NO_2 data for the 2005–2019 pollution window (7). Fourth, we use ACAG SatPM $PM_{2.5}$ as a longer-window pollution extension. Fifth, GADM ADM2 boundaries provide the common subnational geography (8).

Treatment is defined at the district-year level. A district is treated when it is the birth region of the national leader in office in that year. To ensure a unique treatment assignment in transition years, we truncate the earlier of any two consecutive leader spells that overlap within a country.

The core estimating equation uses ADM2 fixed effects and

Significance Statement

Regional favoritism is a well-known finding in political economy: leaders tend to direct economic activity toward their birth regions. This paper asks whether that directed growth is also environmentally cleaner. Using global subnational panels of political leaders, nightlights, and satellite-based pollution, we do not find evidence of cleaner birth-region growth in the later period where pollution data are available. Instead, the central empirical result is that the classic nightlights activity reduced form weakens sharply after 2005. That attenuation limits strong claims about green favoritism and shifts the contribution toward diagnosing how a canonical favoritism result changes in the later satellite era.

R.S. conceived the study, built and integrated the main data pipeline, ran the core empirical analysis, and drafted the manuscript. M.M.A. contributed to the NO_2 analysis. R.W. contributed to manuscript review and proofreading.

The authors declare no competing interests.

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country-year fixed effects:

$$\log(Y_{ict} + 0.01) = \beta\, BirthRegionLeader_{ict} + \alpha_i + \gamma_{ct} + \varepsilon_{ict},$$

where i indexes ADM2 regions, c countries, and t years. Standard errors are clustered by leader-period spell. Following the baseline specification in (1), we estimate the specification for lag 0, lag 1, and lag 2 treatment indicators, but focus on lag 0 for the main results as it provides the most direct test of the regional favoritism hypothesis.

Our later-period environmental outcomes are log nightlights, log NO_2, and a pollution-intensity measure defined as $\log(NO_2 + 0.01) - \log(\text{Nightlights} + 0.01)$. Because both component regressions use the same regressor and fixed effects, the intensity coefficient is mechanically a linear combination of the NO_2 and nightlights coefficients ($\beta_{intensity} = \beta_{NO2} - \beta_{lights}$). We use this measure to test whether birth regions receive cleaner growth (negative coefficient) or dirtier growth (positive coefficient) per unit of economic activity. In the $PM_{2.5}$ extension, the analogous outcome is $\log(PM_{2.5} + 0.01) - \log(\text{Nightlights} + 0.01)$.

2. Activity Reduced-Form Replication and Attenuation

Replication in the long DMSP era. The long DMSP replication remains intact. In the replication-style 1992–2013 sample, the lag-0 birth-region coefficient is 0.0128 with $p = 0.030$. This corresponds to a roughly 1.3% increase in nightlights in treated birth regions, consistent with the magnitudes found in the original regional-favoritism literature (1, 9). The same coefficient is larger in 1995–2013 (0.0187, $p = 0.001$) and remains positive, though weaker, in 2000–2013 (0.0110, $p = 0.056$). These results confirm that the baseline activity effect is not just a narrow late-1990s artifact.

Attenuation in later windows. The key change is that the nightlights reduced form collapses in later windows. In 2005–2013, still within the DMSP era, the lag-0 coefficient falls to 0.0017 with $p = 0.768$. In the harmonized 2005–2019 sample, the lag-0 coefficient is 0.0004 with $p = 0.966$. That pattern makes it difficult to argue that the environmental exercise simply inherits a strong replicated favoritism effect from the classic literature.

The window-grid diagnostics in Figure 1 support that interpretation. Positive coefficients are concentrated in DMSP-only windows with earlier starts and end years around the late 2000s or early 2010s. By contrast, harmonized post-2005 windows remain small and statistically uninformative throughout. The reduced-form problem is therefore not just one bad endpoint choice; it is a broader later-period attenuation pattern.

Additional diagnostics on later-period attenuation. Additional diagnostics in Table 4 narrow the plausible explanations for that attenuation. First, the null is not obviously a pure harmonization artifact. A VIIRS-only check on the standalone 2018–2023 panel is also near zero: the lag-0 coefficient is 0.0038 with $p = 0.722$. Second, the political composition of the sample does not shift enough to make a simple "later years are much more democratic" explanation especially convincing. The share of leader-years in the strict-autocracy bin falls only modestly between the long and later windows, while the more visible change is shorter observed spell support: average active years per spell fall from about 5.22 in 1992–2013 to 3.97 in 2005–2013. We note that the spell-length comparison is partly mechanical due to the shorter 2005–2013 window width, which truncates observed active years for spells starting early or ending late.

Third, treated districts in the later DMSP years are already much brighter than untreated districts. In 2005–2013, the median treated district has DMSP luminosity around 16.84, compared with 8.83 for untreated districts, and the treated upper tail sits close to the DMSP ceiling. That headroom pattern is consistent with mechanical attenuation in later years if leaders are increasingly born in already-bright districts. Supporting that interpretation, the average ascension-year within-country light rank of birth districts is high in both periods and slightly higher in the later sample. By contrast, treatment geocoding quality is stable across periods, so worsening PLAD birthplace measurement is not the leading explanation.

3. Searching for Subsamples Where Favoritism Survives

If the pooled post-2005 activity reduced form masks heterogeneity, the most plausible places to look are settings where the original favoritism mechanism should be strongest or easiest to measure. We therefore test four families of subsamples: geography, political capacity, measurement headroom, and pollution-support samples. Geography captures the fact that Hodler and Raschky find especially strong effects in Sub-Saharan Africa. Political capacity isolates long-tenure leaders and long-tenure autocrats. The headroom splits ask whether DMSP saturation hides effects in already-bright districts. The pollution-support splits ask whether environmental outcomes are only informative in countries with enough NO_2 variation.

This exercise produces one important historical result but no post-2005 rescue. In the full 1992–2013 DMSP window, the Africa coefficient is 0.0428 ($p = 0.009$) and the Sub-Saharan Africa coefficient is 0.0442 ($p = 0.011$), while the non-SSA coefficient is small and insignificant. This confirms that the long-run replication is concentrated in the geography emphasized by the original literature. But in 2005–2013, the Africa and SSA estimates turn negative and insignificant. The other theory-favored post-2005 samples in Table 2 also fail to revive the activity reduced form: long-tenure autocrats are positive but very imprecise, dropping high-luminosity districts does not help, and the high-NO_2 country sample remains near zero.

The harmonized 2005–2019 panel gives the most suggestive but still inconclusive later pattern. Africa (0.0262, $p = 0.167$) and Sub-Saharan Africa (0.0315, $p = 0.135$) are larger than the pooled null, but neither satisfies the activity-effect criterion of a positive coefficient above 0.005 with $p < 0.10$. We therefore treat pollution outcomes in these samples as diagnostic rather than confirmatory (Table 3).

4. Pollution-Side Results

Pollution outcomes in the 2005–2019 sample. Because the later-period nightlights reduced form is weak, the pollution estimates should be read cautiously. They still matter, however, because they do not point toward cleaner birth-region growth even conditional on that limitation. In the pooled 2005–2019 sample (Table 5), the lag-0 nightlights coefficient is near zero, the lag-0 NO_2 coefficient is positive at 0.0308 ($p = 0.045$), and the lag-0 pollution-intensity outcome is also positive at

Table 1. Lag-0 nightlights activity-effect comparison across key windows

Window	Coef.	SE	p-value	N	Treated obs.	Clusters
1992–2013	0.0128	0.0059	0.030	951,915	2,522	2,723
1995–2013	0.0187	0.0055	0.001	829,988	2,196	2,358
2000–2013	0.0110	0.0058	0.056	615,660	1,647	1,759
2005–2013	0.0017	0.0058	0.768	398,636	1,068	1,186
2005–2019	0.0004	0.0087	0.966	671,399	1,701	1,923

All specifications use ADM2 fixed effects, country-year fixed effects, and leader-period clustering. The 2005–2019 row uses the harmonized DMSP–VIIRS nightlights panel.

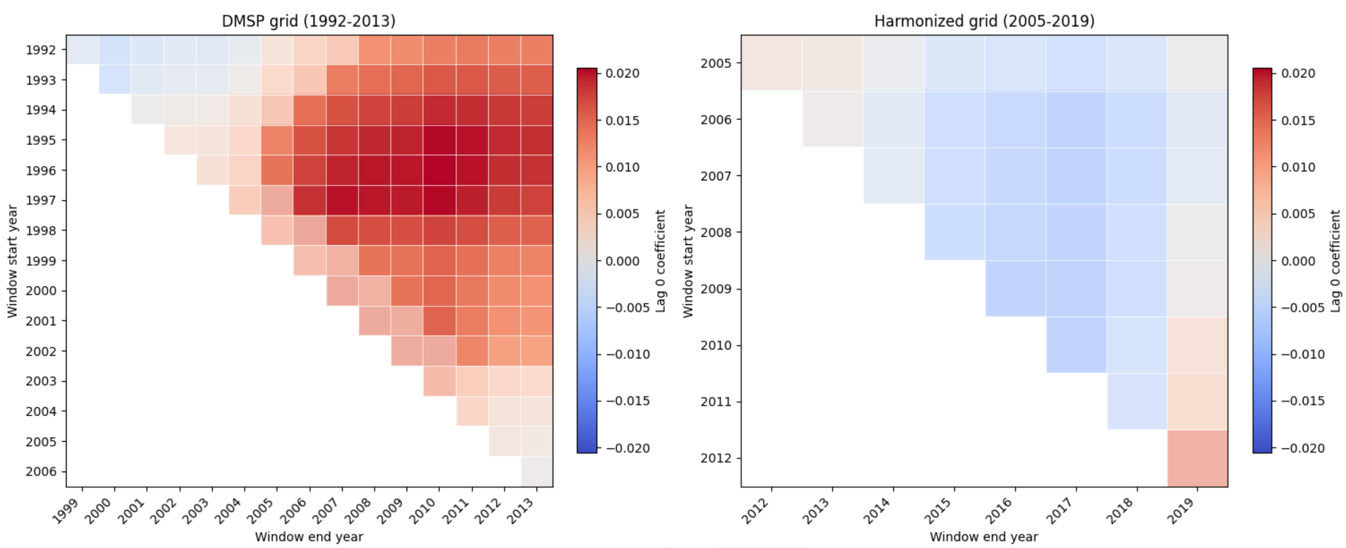

Fig. 1. Lag-0 birth-region nightlights coefficient across all valid start and end windows. Earlier DMSP windows show a block of positive estimates, while post-2005 harmonized windows remain near zero.

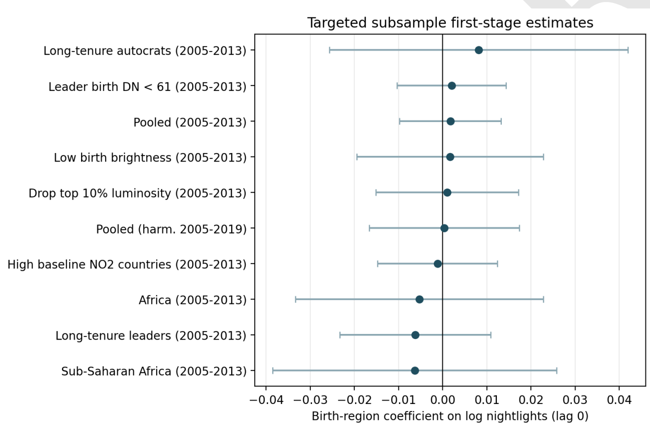

Fig. 2. Targeted post-2005 activity-effect estimates. The plotted samples are the theory-favored cuts from the subsample search, with the pooled harmonized 2005–2019 null included as a reference row.

0.0333 ($p = 0.063$). The implication is not just a lack of green favoritism but rather evidence of dirtier contemporaneous growth in birth regions.

The event-study evidence (not shown) is also not clean enough to rescue a strong causal environmental claim. While positive effects emerge several years after ascension, the pre-period is not flat enough to support a sharp post-treatment interpretation.

Autocracy and $PM_{2.5}$ extensions. The heterogeneity checks do not recover the later activity effect either. Restricting the sample to country-years with $v2x\_polyarchy < 0.3$ yields a lag-0 nightlights coefficient of 0.0111 ($p = 0.574$) in the DMSP 1992–2013 sample and 0.0076 ($p = 0.552$) in the DMSP 2005–2013 sample (Table 6). These estimates are positive but too imprecise to suggest a rescue of the historical result.

The $PM_{2.5}$ extension reinforces the same interpretation. In longer windows, $PM_{2.5}$ level regressions are positive rather than negative, and in the overlapping 2005–2019 sample the $PM_{2.5}$ pollution-intensity outcome is also not negative. The lag-2 estimate in Table 5 is positive and statistically significant, though this should be interpreted cautiously given multiple testing across lags. Taken together, the evidence is more consistent with the absence of green favoritism than with a cleaner-growth interpretation.

5. Discussion

The results do not support the original green-favoritism hypothesis. In the window where pollution analysis is feasible at scale, the nightlights activity effect is close to zero and the pollution estimates are zero to positive rather than negative. While the long DMSP replication remains visible, the later-period environmental extension cannot be interpreted as a simple continuation of the classic result.

Table 2. Targeted activity-effect subsample search

Sample	Window	Coef.	SE	p-value	N	Treated	Countries
Africa	DMSP 1992–2013	0.0428	0.0163	0.009	109,126	705	48
Sub-Saharan Africa	DMSP 1992–2013	0.0442	0.0174	0.011	63,378	631	45
Non-SSA	DMSP 1992–2013	0.0053	0.0059	0.370	888,537	1,891	187
Pooled	DMSP 2005–2013	0.0017	0.0058	0.768	398,636	1,068	235
Africa	DMSP 2005–2013	-0.0052	0.0143	0.714	46,939	306	48
Sub-Saharan Africa	DMSP 2005–2013	-0.0063	0.0164	0.700	28,191	279	45
Long-tenure leaders	DMSP 2005–2013	-0.0062	0.0087	0.479	300,372	862	134
Long-tenure autocrats	DMSP 2005–2013	0.0081	0.0173	0.637	42,717	261	40
Drop top 10% luminosity	DMSP 2005–2013	0.0010	0.0082	0.899	357,745	613	230
Leader birth DN < 61	DMSP 2005–2013	0.0020	0.0063	0.750	373,862	1,034	234
Low birth brightness	DMSP 2005–2013	0.0016	0.0108	0.880	142,320	634	98
High baseline NO_2 countries	DMSP 2005–2013	-0.0012	0.0069	0.866	312,633	603	114
Africa	Harmonized 2005–2019	0.0262	0.0189	0.167	83,101	511	48
Sub-Saharan Africa	Harmonized 2005–2019	0.0315	0.0210	0.135	52,001	466	45
Low NO_2 dispersion countries	Harmonized 2005–2019	0.0219	0.0183	0.232	95,961	621	124

All rows use lag-0 treatment, ADM2 fixed effects, country-year fixed effects, and leader-period clustering. Long tenure means total PLAD spell length of at least five years. Long-tenure autocracy additionally requires $v2x\_polyarchy < 0.3$. The activity-effect criterion is coef. ≥ 0.005, $p < 0.10$, and at least 50 treated observations in a post-2005 sample. No post-2005 subsample satisfies that criterion.

Table 3. Pollution outcomes in strongest positive diagnostic samples

Sample	FS coef.	FS p	NO_2 intensity	NO_2 p	$PM_{2.5}$ intensity	$PM_{2.5}$ p
Sub-Saharan Africa	0.0315	0.135	-0.0323	0.237	-0.0336	0.127
Africa	0.0262	0.167	-0.0267	0.295	-0.0267	0.178
Low NO_2 dispersion	0.0219	0.232	0.0156	0.685	-0.0203	0.281

The activity reduced-form (FS) columns use the harmonized 2005–2019 nightlights panel. Pollution-intensity outcomes are log pollution minus log nightlights in the same subsample. These are diagnostic estimates because none of the activity effects is statistically significant under the stated criterion.

Table 4. Compact diagnostics on the later null

Diagnostic	Reference Group	Treated / Later Group
VIIRS-only lag-0 coef.	—	0.0038
VIIRS-only lag-0 p-value	—	0.722
Avg. active years per spell	5.22	3.97
Median DN, untreated vs treated	8.83	16.84
Treated DN upper tail ($p95$)	—	61.00

The spell-length comparison uses 1992–2013 (Reference) versus 2005–2013 (Later). The luminosity comparison uses the DMSP 2005–2013 sample. DN refers to Digital Number (0–63).

The paper's main contribution is therefore diagnostic. The later-period attenuation appears broad enough that it cannot be dismissed as a single endpoint choice. While we find suggestive patterns regarding shorter observed leader spells and reduced luminosity headroom, these explanations are not fully definitive. Specifically, the subsample tests dropping high-luminosity districts do not recover the effect, and spell-length attenuation is partly mechanical in the shorter window. Moreover, we note that the 95% confidence intervals for the post-2005 coefficients (e.g., Table 1) still contain the historical 1.3% effect, so we cannot statistically rule out a constant-effect null; the results simply indicate a loss of detectable signal.

That framing is also the cleanest way to connect the nightlights and pollution evidence. Rather than claiming cleaner growth in leader birth regions, the current results suggest a later political era in which the activity effect itself is weaker or harder to detect. Once that is true, the absence of cleaner growth is no longer surprising. The strongest later diagnostic samples in Table 3 do show negative pollution-intensity point estimates in Africa and Sub-Saharan Africa, but those estimates are imprecise and cannot carry the original green-favoritism claim.

ACKNOWLEDGMENTS. We thank Professor Joshua Blumenstock for feedback. All errors are our own.

Data, Materials, and Code Availability

Code, notebooks, and manuscript sources are available in the public GitHub repository at https://github.com/RichSchulz/political-leader-pollution (Commit hash: `5d8f2a1`). The analysis reproduces Tables 1–5 via `final_analysis.ipynb`. Data inputs include PLAD, GADM, ACAG, and V-Dem country-year data.

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Table 5. Main later-period pollution results

Outcome	Window / spec	Coef.	SE	p-value	N
$\log(\text{Nightlights} + 0.01)$	2005–2019 lag 0	-0.0026	0.0091	0.778	631,377
$\log(\text{NO}_2 + 0.01)$	2005–2019 lag 0	0.0308	0.0154	0.045	631,377
Pollution intensity	2005–2019 lag 0	0.0333	0.0179	0.063	631,377
$\text{PM}_{2.5}$ pollution intensity	2005–2019 lag 0	0.0041	0.0095	0.665	669,438
$\text{PM}_{2.5}$ pollution intensity	2005–2019 lag 1	0.0122	0.0078	0.120	669,438
$\text{PM}_{2.5}$ pollution intensity	2005–2019 lag 2	0.0175	0.0074	0.019	669,438

Pollution intensity is $\log(Pollution + 0.01) - \log(Nightlights + 0.01)$. Negative coefficients would indicate cleaner growth in leader birth regions; current estimates are zero to positive.

Table 6. Appendix-style autocracy reduced-form check

Window	Coef.	SE	p-value
1992–2013 autocracy lag 0	0.0111	0.0197	0.574
2005–2013 autocracy lag 0	0.0076	0.0127	0.552

Autocracy is defined as $v2x\_polyarchy < 0.3$.

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